exact equations

a diffrential expression $M(x,y)\,dx + N(x,y)\,dy$ is an exact diffrential equation must look like:

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

exact equation means: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ the partial derivative of M with respect to y must equal the partial derivative of N with respect to x

so before you start any problem you need to make sure the partial derivatives are vaild

example:

1) $\underbrace{2xy}_{M}\,dx + \underbrace{x^2}_{N}\,dy = 0$

$$\left.\begin{aligned} M &= 2xy \quad\Rightarrow\quad \frac{\partial M}{\partial y} = 2x \\ N &= x^2 \quad\Rightarrow\quad \frac{\partial N}{\partial x} = 2x \end{aligned}\right\} \text{exact } \checkmark$$

when they match you can continue with the process of solving if the partial derivatives match you can continue with the problem solving if the partial derivatives don't match you cannot continue with the problem solving

when taking the integral $M(x,y)$ or M needs to be associated with dx and $N(x,y)$ or N needs to be associated with dy

Practice problems

1) $\underbrace{(x+y)^2}_{m} dx + \underbrace{(2xy + x^2 - 1)}_{n} dy = 0$

$M = (x+y)^2 \Rightarrow x^2 + 2xy + y^2 \Rightarrow \frac{\partial m}{\partial y} = 2x + y$

$n = 2xy + x^2 - 1 \Rightarrow \frac{\partial n}{\partial y} = 2x + 2y$

• the partial derivatives give us the same answer for both of them

$\int m\, dx \Rightarrow \int (x^2 + 2xy + y^2)\, dx = \frac{1}{3}x^3 + x^2y + xy^2 + c_1$

$\int n\, dy \Rightarrow \int (2xy + x^2 - 1)\, dy = xy^2 + x^2y - y + c_1$

when you are finding the general solution you copy both solutions into one but if there is the same term in both equation you never Repeat it you only copy it down into the general solution once then set it equal to a constant

gen soln $\Rightarrow$ $\boxed{\frac{1}{3}x^3 + x^2y + xy^2 - y = C}$

notice x^2y repeated it self in each solution of the integral for m and n but we included it once in the general solution

2) $\underbrace{(-3x^2-y)}_{m}\,dx + \underbrace{(3y^2-x)}_{n}\,dy = 0$

$m = -3x^2-y \Rightarrow \frac{\partial m}{\partial y} = -1$

$n = 3y^2-x \Rightarrow \frac{\partial n}{\partial x} = -1$

• the Partial derivatives For m and n give us the same exact solution which means we can continue solving

$\int m\,dx \Rightarrow \int (-3x^2-y)\,dx = -x^3-xy+c_1$

$\int n\,dx \Rightarrow \int (3y^2-x)\,dy = y^3-xy+c_2$

gen Soln $\Rightarrow \boxed{-x^3-xy+y^3 = c}$

notice xy was repeated twice in the integral solution of m and n but we only included it once in the general solution

3) $\left(1+\ln(x)+\frac{y}{x}\right)dx = \left(1-\ln(x)\right)dy$

begin by getting this into Standard Form then you can use Partial derivatives to check if they are exact

$$\underbrace{\left(1+\ln(x)+\frac{y}{x}\right)}_{m}dx+\underbrace{(\ln(x)-1)}_{n}dy=0$$

$m = 1+\ln(x)+\frac{y}{x} \Rightarrow \frac{\partial m}{\partial y} = \frac{1}{x}$

$n = \ln(x)-1 \Rightarrow \frac{\partial n}{\partial x} = \frac{1}{x}$

- these two are exact so you can continue solving the DE

$$\int m\,dx \Rightarrow \int \left(1+\ln(x)+\frac{y}{x}\right)dx \Rightarrow \int 1\,dx+\int \ln(x)\,dx+\int \frac{y}{x}\,dx$$

$$\int 1\,dx = x$$

$$\int \ln(x)\,dx = x\ln(x)-x$$

you would solve this by integration by parts $u=\ln(x)$ $dv=dx$ but just memorize it

$$\int \frac{y}{x}\,dx = y\ln(x)$$

$$\int m\,dx = \cancel{x}+x\ln(x)-\cancel{x}+y\ln(x)+C \Rightarrow x\ln(x)+y\ln(x)+C$$

$$\int n\,dy = \int (\ln(x)-1)\,dy \Rightarrow y\ln(x)-y+C_2$$

$$\boxed{x\ln(x)+y\ln(x)-y = C}$$

notice how $y\ln(x)$ repeated in both integral solution but we only included it once it only once

4) $x \frac{dy}{dx} = 3xe^x - y + 9x^2$

• begin by getting this into standard form

$$x\,dy = (3xe^x - y + 9x^2)\,dx$$

$$x\,dy - (3xe^x - y + 9x^2)\,dx = 0$$

$$\underbrace{x\,dy}_{n} + (\underbrace{-3xe^x + y - 9x^2}_{m})\,dx = 0$$

remember n needs to be associated with dy and m needs to be associated with dx

$n = x \Rightarrow \frac{\partial n}{\partial x} = 1$

$m = -3xe^x + y - 9x^2 \Rightarrow \frac{\partial m}{\partial y} = 1$

• both of these are exact so you can continue solving the DE

$\int n\,dy \Rightarrow \int x\,dy \Rightarrow xy + c_1$

integration by parts

$\int m\,dx \Rightarrow \int (-3xe^x + y - 9x^2)\,dx \Rightarrow \int -3xe^x\,dx + \int y\,dx - \int 9x^2\,dx$

$uv - \int v\,du$

lipet

$u = -3x$

$du = -3\,dx$

$dv = e^x$

$v = \int e^x\,dx$

$v = e^x$

$uv - \int v\,du$

$-3xe^x + 3\int e^x\,dx$

$-3xe^x + 3e^x$

$$-3xe^x + 3e^x + xy - 3x^3 = c_2$$

$$\boxed{-3xe^{x} + 3e^{x} + xy - 3x^{3} = C}$$

notice how xy was repeating in both the integral solution but we only copied it down once for the final solution

$$5)\ \underbrace{(x+y)^2}_{m} dx + (\underbrace{2xy + x^2 - 3}_{n})\, dy = 0$$

$m = (x+y)^2 \Rightarrow (x+y)(x+y) \rightarrow x^2 + 2xy + y^2 \Rightarrow \dfrac{\partial m}{\partial y} = 2x + 2y$

$n = 2xy + x^2 - 3 \Rightarrow \dfrac{\partial n}{\partial x} = 2y + 2x$

- these are exactly the same so you can continue solving the DE

$\int m\, dx \Rightarrow \int (x^2 + 2xy + y^2)\, dx \Rightarrow \dfrac{x^3}{3} + x^2y + xy^2 + c_1$

$\int n\, dy \Rightarrow \int (2xy + x^2 - 3)\, dy \Rightarrow xy^2 + x^2y - 3y + c_2$

$$\boxed{\frac{x^3}{3} + x^2y + xy^2 - 3y = C}$$

notice how xy^2 and x^2y are repeating in the integral solution but we only copied them once for the final equation